Alan Chung

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Education:

- **Harvard University PhD Candidate:** Department of Statistics, focusing on Probability Theory and Theoretical ML
- **Princeton University Class of 2022:** AB Mathematics, focusing on Probability Theory and Statistical Learning Theory
  - **GPA**: 3.96 Cumulative, 3.98 Technical
  - **Sample of Relevant Coursework:** Real and Complex Analysis, Probability Theory, Abstract Algebra, Programming Systems, Stochastic Calculus, Combinatorics, Reinforcement Learning, Quantum Mechanics, High-Dimensional Probability, Statistical Physics. (Prospective Spring 2022: Functional Analysis, Advanced Statistical Physics)
  - **Self-Studied Courses:** COS511 (Online Convex Optimization)
  - **Organized** a Quantum Cryptography Reading Group Winter 2020 under Professor **Mark Zhandry**
- **Scarsdale High School:** Graduated June 2018. GPA 4.18/4.0 Unweighted

Research Experience: Probability Theory and Statistical Learning Theory

- **June 2021 – Present:** Work in **Stochastic Calculus/PDEs** with Professor **Mykhaylo Shkolnikov of Princeton University**
  - Working to model Melting Patterns in Polar Ice Caps using tools in Stochastic Calculus. We are investigating melting rates by using a probabilistic formulation to study this free boundary PDE problem.
- **January 2021 – Present:** Work in **Theoretical Reinforcement Learning** with Professor **Chi Jin of Princeton University**
  - Worked to extend the results in this paper on Learning Undercomplete POMDPs to the Overcomplete case. We have devised a Maximum Likelihood Approach (MLE) to Learning Overcomplete POMDPs in the Exploration setting and are writing the results in the **working paper**.
- **June – August 2020:** Work in **MCMC methods**, with Professor **Ryan Adams of Princeton University**
  - We devised a Gibbs Sampling Algorithm for the Birkhoff Polytope, which is the space of N dimensional doubly stochastic matrices. We sought to prove convergence rates for this algorithm.
- **January – June 2020:** Work in **Gaussian Processes**, with Professor **Ryan Adams of Princeton University**
  - Attempting to analyze and bound the distribution of local extrema of Gaussian Processes. This project is motivated by the loss surfaces of Deep Neural Networks
- **June – August 2019:** Work in **Geometric Algebra**, with Professor **Ryan Adams of Princeton University**
  - Investigating the application of Geometric Algebra in ML. We found a compact formula for a high-dimensional radical center using the Geometric Algebra.
- **June – August 2017:** Work in **NLP** with Professors **Luis Gravano and Daniel Hsu of Columbia University**
  - Use ML Tools to identify food poisoning cases from customers' reviews on sites such as Yelp, TripAdvisor, etc. This project was in conjunction with the NYC Health Department as part of a public-health initiative in NYC.

Work Experience

- **June – August 2021:** **Quantitative Research Intern** at **Citadel Securities**
  - Engaged in Alpha Research; Investigated the Relationship Between the Liquid/Illiquid Hours in the Futures Market
- **August – December 2020:** **Grader** for COS324 (Introduction to Machine Learning)
- **February 2020 – May 2021:** **Teaching Assistant** for COS 302 (Mathematics for Numerical Computing and ML)
- **October – August 2019:** **Problem Writer** for PUMaC (Princeton University Mathematics Competition)
- **June – August 2018:** **Teaching Assistant** at AwesomeMath Summer Mathematical Olympiad Training Program

Academic Honors

- **USAPhO** (United States Physics Olympiad) Honorable Mention
- **USAMO** (United States Mathematical Olympiad) Qualifier
- **USACO** (United States Computing Olympiad) Gold Division Qualifier
- **NYSML** (New York State Mathematics League) Top Ten Individual

Programming, Mathematics Skills, and Extracurricular Activites

- Proficient in Java, Python, C++, MATLAB, Linux/Unix, and LaTeX.
- Manuscript Containing Notes/Solutions to Difficult Math and Physics Olympiad Problems Here: **Link**
- 2020: Advising Chair for the **Princeton University Math Club.**
- 2015 – 2018: Member of the Princeton Competitive Club Tennis Team.
- Language: Native English; Fluent Mandarin, Proficient Spanish
- Age of Empires II Enthusiast